CHAPTER SIX
Corporations and sustainability

6.1 Development's Risks

6.1.1 Manufactured risk

On the night of 2 December 1984, an explosion in a pesticide factory in Bhopal in Madhya Pradesh, India, spread a cloud of pollution over the city. The plant, owned by Union Carbide India Limited (in turn 50.9% owned by the Union Carbide Corporation of the USA), was built on the outskirts of Bhopal city, 1 kilometre from the railway station and 3 kilometres from two hospitals. Over the next hour, about 30 metric tons of methyl isocyanate gas escaped and sank over the surrounding streets. Within two days, five thousand people had died from inhaling the gas, and half the population of the city had fled in terror. The eventual total death toll has been estimated to be as high as 20,000 people, with hundreds of thousands of people suffering chronic poisoning (Low and Gleeson 1998, Varma and Varma 2005, Narain and Bushan 2015).

The accident in Bhopal happened when a tank was being cleaned. Water came into contact with liquid methyl isocyanate and generated gas that vented to the atmosphere. Responsibility for the tragedy remains a moot point: the explosion might have been averted had the automatic controls standard in such Union Carbide plants in the USA been installed, had the Indian government not insisted that manual controls be fitted, or had employees of the Indian subsidiary of Union Carbide followed plans more closely. The human cost of the disaster would have been much less if the plant had not been built so close to the city. The Indian government made an out-of-court settlement with Union Carbide in 1989 for $470 million, although legal actions continued, both to make the state pay out this compensation to victims and to make the plant's new owners (Dow Chemicals) clean up pollution (Varma and Varma 2005). By 2009, the compensation had grown to $1.9 billion (still a seventh of the Indian government's original demand). In return, all civil and criminal cases against the company were terminated (Narain and Bushan 2015). At the start of legal proceedings, the government said there had been some 3,000 deaths and 30,000 cases of injury. By then it had become clear that huge numbers of people were suffering from gas exposure, and over half a million people received small amounts of compensation (5,295 death cases and 568,293 classified as injured). After the disaster, 350 tonnes of toxic waste was kept in a leaking shed at the site, polluting soil and groundwater and affecting the health of the people of Bhopal (Narain and Bushan 2015).

Fifteen months after the Bhopal disaster, the German sociologist Ulrich Beck published his book *Risikogesellschaft* (published in English in 1992 as *Risk Society*). Beck set out the idea that there was a break within modernity, a transition from classical industrial society towards a new (but still industrial) form of 'risk society'. In place of localised risks, to nature, employees or local people, new forms of techno-scientific risk offered a global threat. As he put it, 'a new twilight of opportunities and hazards comes into existence' (Beck 1992, p. 15). Beck's arguments picked up on long-standing environmentalist concern that science and technology were producing toxic materials and life-threatening processes that were novel in their longevity, the scale of their impact, their invisibility and their complexity. He suggested that

'social, political, economic and individual risks increasingly tend to escape the institutions for monitoring and protection' (Beck 1992, p. 5). Global ecological crisis could, therefore, no longer be thought of in terms of local 'environmental problems' but needed instead to be understood as 'a profound institutional crisis of industrial society itself' (Beck 1992, p. 8).

Beck defined risk as 'the probabilities of physical harm due to given technological or other processes' (Beck 1992, p. 4). The risks of pesticide production at Bhopal were, to Beck, part of 'a bargain struck by the Indian state on behalf of its people', a deal that balanced risk of poisoning and death against the benefits of work and useful agricultural fertilisers, pesticides and herbicides. The plant itself, and its belching pipes and steaming tanks, were an expensive symbol of developmental success, their products signifying 'emancipation from material need'. By contrast, the 'death threat' they contained remained largely invisible (Beck 1992, p. 42).

The pesticides being produced by Union Carbide in Bhopal were carbamates, including carbaryl (sold under the trade name Sevin), aldicarb (trade name Temik) and a mix of carbaryl and the organochlorine pesticide lindane, sold under the trade name Sevidol (Narain and Bushan 2015). All were part of the package of technical inputs of the 'Green Revolution', intended to revolutionise agricultural productivity in India as elsewhere and push back hunger and poverty (see Chapter 10). There is irony in 'the competition between the visible threat from hunger and the invisible threat of death from toxic chemicals' (Beck 1992, p. 42).

This choice between these unlike abstractions of promise and risk is inherent to thinking about development and sustainability. Economic growth is regarded as necessary to reduce human poverty (Chapter 3), and industrialisation under capitalism is taken to be a prerequisite of economic growth (Chapter 5). That is how the earth's currently industrialised (and wealthy) core countries got where they are, and copying them is the one big idea of the dominant developmentalist paradigm.

Development is a two-edged sword, promising to hack away at the choking creepers of poverty but at the same time bringing with it unrecognised, unregulated and often deeply hazardous change. Furthermore, the risks development creates are not distributed uniformly but are concentrated in space and time. They are faced disproportionately by the poor. On top of the burdens of ill health and poverty, they bear the largest hazards of technology's products in their workplace (if they can find work) and in the degraded and unhealthy environmental conditions where they live. Limited governance capacity conspires with the naked urge of capital to treat the environment and the bodies of the poor as externalities, exposed to the hard edge of techno-scientific risk.

As discussed in Chapter 5, the logic of capitalism drives industry to relocate in search of cheap labour and resources and the freedom to externalise the costs of production. At the centre of that process lies the corporation, an institution structured to generate profit for investors. As the cutting blade at the sharp edge of capitalism, corporations have long sought profit in the global periphery, ever since the days of the giant imperial trading houses such as the British East India Company, ruthlessly annexing and ruling foreign lands in pursuit of revenue. The general profitability of industrialisation in developing countries was a phenomenon of the late twentieth century (Marshall 1999). It was possible because of changes in infrastructure, economy and society in developing countries, increased global economic integration, a revolution in communications and the mobility of capital.

The global capitalist system thrives by externalising the costs of environmental degradation (Gadgil and Guha 1995). Low costs, cheap labour and limited environmental regulations provide opportunities to maximise profit in industrial ventures in the developing world. By the end of the twentieth century, industrialised countries had begun to evade the risks they had conjured up with their industrial technology by moving them elsewhere. One effect of the search for sustainable development in industrialised countries has been the export of unsustainability offshore. Even as sustainable development became part of the language of developed-country governments, their citizens were importing the products of new factories in new industrial zones on the global periphery. The gleaming products that feed the West's insatiable consumer demand show little or nothing of how they are made. Child labour,

sweatshops, industrial pollution and urban poverty are far away down long chains of trade that are only slowly being made visible.

But the Bhopal disaster not only reflects a state of technology and modernity but also the nature of capitalism. Bhopal was brought about by the decisions within a private corporation. It takes its place in a long history of industrial accidents. The hazards of dangerous, poorly understood and unregulated industrial processes have long offered a threat to the welfare of workers – as, for example, in the notorious case of the employees of the US Radium Corporation in the 1920s, who contracted mouth cancer licking the brushes with which they put luminescent dots of radioactive paint on watch faces (Caufield 1989).

As industrial processes are scaled up, so too is the magnitude of accidents. There have been extensive industrial disasters before in industrialised countries that played a role in the development of environmentalism; for example, the fire at the first nuclear reactor in the UK at Sellafield (then Windscale) in the UK in October 1957; the explosion at Flixburgh in the UK in 1974; the release of dioxin at Seveso in Italy in 1976; and the discovery of a toxic waste dump from the 1950s at Love Canal in New York State in 1978. Many other disasters have followed, such as the meltdown of the nuclear reactor at Chernobyl in the Ukraine in 1986; the explosion on the Piper Alpha oil rig in the North Sea in 1988; the Exxon Valdez oil spill in Alaska in 1989; the fire at the Texas City oil refinery in the USA in 2005; the Deepwater Horizon drilling rig explosion in the Gulf of Mexico in 2010; or the fire at the Tasreen Fashions factory in Dhaka in Bangladesh in 2012. Each corporate industrial disaster has its short moment of media attention but is soon forgotten. The catalogue of risk and hazard is long and shocking.

6.1.2 The politics of risk

Ulrich Beck's concept of 'risk society' (discussed above) drew attention to modernity's difficulties with finding rational solutions to the environmental hazards that they have brought into being (Matten 2004). He is clearly right that the modern techno-economic system produces risks at the same time as it produces wealth (Beck 1992, 1994). The risks of modernisation are 'a *wholesale product* of industrialization and are systematically intensified as it becomes global' (Beck 1992, p. 21; emphasis in the original). He is surely also right that the novel form and scale of industrial risks (for example, the long-term health effects of complex organic compounds or the trans-boundary pollution risks of nuclear industrialisation) make it hard for conventional governance institutions to manage them. Techno-scientific risks are not readily calculable and are not limited by traditional boundaries of location or time span or social groups (Matten 2004).

What is more controversial is Beck's further suggestion that risks are no longer limited to particular social groups but are, potentially at least, universal. The threat of late modern risks is so enormous that they cut across class lines. So even those who profit from the techno-scientific risks that modernity brings will eventually be affected (the risks of a major nuclear accident being hard for even the rich and powerful to avoid, for example). Here Beck's ideas run against experience of the political ecology of risk. Risk is translated through socio-economic status, geographical location, cultural values and belief (Mythen 2005). Not everybody experiences the risks of modernisation in the same way, or to the same extent. Hazards are not universal. They reinforce existing class divisions rather than transcend them (Marshall 1999). As Harvey (1993, p. 27) put it

> not only do the rich occupy privileged niches in the habitat while the poor tend to work and live in the more toxic or hazardous zones ... but the very design of the transformed ecosystem is redolent of its social relations.

The impact of risk therefore remains fundamentally affected by social stratification. The idea of techno-scientific risk as a boomerang that comes back to afflict the societies that create it may be true, but some societies (in the developed world) and some classes of people (those who can move, get treatment and invest to protect themselves) are much less vulnerable than others: as Mythen (2005, p. 141) puts it, 'the boomerang effect may bruise some, but it will administer knock-out blows to others'. Beck's risk society thesis fails to make the case that risk effects can be separated from class.

There is therefore a distributional logic to risk, both between rich and poor in industrialised countries, and internationally between developed and developing countries. There is an associated danger that the pursuit of development will lead to neglect of environmental justice (Merryfield and Swyngedouw 1996, Low and Gleeson 1998). In economic terms, it is logical to locate hazardous pollution where costs are low, where the poor are willing to accept low levels of compensation. But this outcome is unjust. Community leaders are faced with the dilemma of 'trading their people's environmental health in return for basic material security' (Low and Gleeson 1998, p. 119). Debate in the USA drew attention to the differential exposure of minorities, native people and people of colour to environmental risks, particularly in the location of toxic waste dumps and other polluting facilities. The manifest unfairness of the outcome of corporate and municipal decisions about noxious facilities gave rise to the environmental justice movement and the concept of 'environmental racism' (Harvey 1996).

Environmental justice was an important element in wider concern about environmental health, a key element in the growth of environmentalism in the USA (Hays 1987). The movement has been built around particular cases, many of them highly disturbing and high profile. One such was the protest about Love Canal in Buffalo, New York, in 1977, where, between 1942 and 1952, the Hooker Chemical and Plastics Corporation buried about 22,000 tons of toxic waste. Subsequently, a school and houses were built on the landfill. By the late 1970s, severe health problems were being experienced in the area, and children were being born with severe disabilities. Following protest, a Federal State of Emergency was declared, and residents were rehoused. The Environmental Protection Agency sued Occidental Petroleum (which had taken over the Hooker company), who paid a substantial sum in restitution in 1995.

Love Canal was followed by many other protests in the 1980s and 1990s about the location of hazardous facilities and threats to human health and wellbeing – for example the opposition to the proposed Brooklyn Navy Yard incinerator in New York (Gandy 2002). By the 1990s, the environmental justice movement was well established in the USA, with strong offshoots elsewhere (Harvey 1996). In response, the US Environmental Protection Agency established an Office of Environmental Equity in 1992, and in 1996 an Executive Order required every federal agency to consider the effects of its programmes on the health and wellbeing of minority communities (Low and Gleeson 1998). Environmental justice has also been an important focus of the environmental movement in developing countries, often inspired by campaigns against corporate projects such as dams, mines, factories and oilfields (Martínez-Alier *et al.* 2016, Temper *et al.* 2018b, see also section 6.4 below and Chapter 13).

6.1.3 Regulating hazard

International trade creates a wider geographical scale for the distributional logic of industrial hazard and pollution. Just as the logic of capitalism at the urban scale leads to the location of noxious facilities in poor districts, so the same logic at the global scale dictates that the most efficient logical place to locate noxious industrial processes is not in the industrial world's increasingly clean and self-proclaimed 'sustainable' Northern countries but at the feet of the poor in developing countries. There, determination to overcome poverty and lax environmental regulations attract hazardous industries 'like magnets' and create an explosive mixture. As Ulrich Beck put it, 'the devil of hunger is fought with the Beelzebub of multiplying risks' (Beck 1992, p. 43). In the face of globalised corporate power, the nation state (particularly in the developing world) is often a weak player, unable to regulate environmental risk.

Developing countries often lack the environmental and anti-pollution safeguards that have become standard in the industrialised world. Walter and Ugelow (1979) showed an inverse relationship between level of development and rigour of environmental policies in 145 countries, and this pattern persists. Lack of pollution controls cuts the cost of production, or rather transfers costs to the host environment and community. In a sense, therefore, pollution is a hidden subsidy for developing-world industry. Developing countries with a suitable and cheap workforce and undemanding employment law can

achieve a competitive advantage over industrialised countries as destinations for many manufacturing enterprises (Walter and Ugelow 1979).

There is certainly evidence that, in the past, lax pollution controls in the developing world attracted corporate investment. US-based multinationals in primary metals industries spent 21 per cent of capital on pollution control in their US operations but less than 10 per cent abroad (Walter and Ugelow 1979). Proportions were similar for other industries; for example, chemicals (9% in the USA, 5% abroad). Some countries (like South Africa) seem to constitute 'pollution havens' (Nahman and Antrobus 2005). In 1986, the British company Thor Chemicals began importing waste from the USA and Europe to its reprocessing plant in South Africa. By 1988, mercury levels 1,000 times the WHO standards were discovered in a river 50 kilometres away, and by 1992 workers had started to die in the plant, and others had become disabled. The plant closed in 1994, but the pollution remained.

The movement of polluting industrial activity to developing countries has long been directly linked to the rise of pollution control in industrialised countries. In 1966, a Japanese company built a plant making caustic soda on the coast of Thailand (Suckcharoen *et al.* 1978). The decision to locate this factory, and its methyl mercury pollution, outside Japan was attractive because Thai environmental protection legislation was weak; in Japan, tight regulations had been passed following acute organic mercury poisoning discovered among the fishing communities in Minamata Bay, Kyushu Island, in the 1950s. This was the result of pollution from the Chisso Corporation's fertiliser factory (Vlachou 2004), which caused extensive birth defects in children and sickness and mortality in adults ('Minamata disease'). This incident was important in the rise of the environmental movement in Japan, and it raised awareness of the dangers of industrial pollution globally (D'Itri and D'Itri 1977, Avenell 2013). The Minamata Convention on Mercury, agreed in 2013 and named after the Japanese town, seeks to control all anthropogenic emissions and releases of mercury and mercury compounds and ban its use in manufacture, import and export by 2020 (Minamata Convention on Mercury 2019).

A particular example of this link between the environmental concerns in developed and developing economies is the burgeoning international trade in waste, particularly toxic and hazardous waste. In the 1990s, World Bank economists made for the economic efficiency of locating toxic and hazardous waste dumps in poor countries (Rich 1994, Harvey 1996). Today, there is a growing industry in breaking ships and reprocessing shipped electrical waste in developing countries. Mobile phones, computers, fridges and other consumer devices, many of them rendered obsolete by technical advance and the relentless dictates of fashion, are dispatched by the container load to the developing world. As long ago as 2002 the USA was exporting over ten million obsolete computers (Iles 2004). Working conditions in these industries are very poor, with workers exposed to hazardous compounds and heavy metals without training, protective clothing or any form of social insurance.

There is also a trade in gathered toxic waste (rather than just waste products with toxic elements). International trade in hazardous waste is covered by the 1989 Basel Convention on the Control of Transboundary Movements of Hazardous Wastes, and was addressed in *Agenda* 21. These agreements reflected shocking cases of hazardous waste dumping in the developing world. In 1988, Guinea-Bissau was offered a contract for $600 million (four times its GNP) to dispose of 15 million tons of toxic waste over five years (Lipman 2002). That deal did not go though, partly because environmentalists opposing it were able to draw on warnings from a similar dump attempt in nearby Nigeria. In 1988, Greenpeace revealed that two Italian corporations had paid a businessman in the town of Koko in the Niger Delta in Nigeria about $100 per month to store 18,000 drums of hazardous wastes disguised as building materials (including asbestos, dimethyl formaldehyde and PCBs). Workers clearing the site suffered illness, and the town has been virtually abandoned (Adeola 2001, Lipman 2002). Meanwhile, the ship the *Karin B* toured the ports of Europe in search of a place to unload the toxic waste where it could be properly processed.

The 1991 Bamako Convention subsequently banned waste imports to Africa, and in 1994 the parties to the Basel Convention agreed to ban all export of hazardous waste from countries in the Organisation for Economic Co-operation and Development (OECD) (that is, developed countries) to non-OECD countries. Yet international transfers of hazardous waste remain standard. Trade in hazardous as well as

non-hazardous waste and scrap has continued to grow rapidly in the current century, driven by international differentials in the stringency of environmental policy, efficiency of governance, rates of taxation and transport costs (Kellenberg 2015). A substantial part of the trade is illegal, and organised crime plays a role in some countries. The trade, and the lack of regulation along the disposal chain, has serious implications for the welfare and health of workers and the state of the environment.

In accepting industrial investment even though their regulatory capacity and governance are weak, developing countries are effectively trading off hopes of future national economic security against present risk. In these exchanges, 'sustainability' within one jurisdiction (the industrial country where corporations are domiciled) is being achieved by exporting risk and hazard elsewhere. Yet the movement of polluting industries to the developing world is supported on all sides, by governments, international banks and financial houses, and global corporations. The 'traffic in risk' (Low and Gleeson 1998) may be attractive to both developing-country governments eager for investment and industrialised country governments, who are able to pacify environmentalists at home by moving hazardous elements in the production process outside their borders. Lack of facilities for identifying or measuring environmental hazards and costs means they are often underestimated or ignored. Long trans-global production chains tend to restrict transparency about the conditions of production. It can seem profitable to 'manipulate and bribe politicians and bureaucrats rather than worry about technological innovation, efficient resource use or pollution control' (Gadgil and Guha 1995, p. 31).

Manufacturers often have a good idea of the dangers implicit in their operations, although they seek to shield this knowledge from their consumers, environmental organisations and sometimes from the state, in order not to jeopardise their search for profit (Murphy 1994). Environmental and social impacts, therefore, are in a sense not unexpected but events whose likelihood of occurrence is carefully calculated by corporate planners. Techno-scientific risk is an integral element in the evolution of industrial society.

6.2 Greening Business

6.2.1 Environmentalisms versus the corporation

These arguments about corporate responsibility for techno-scientific risk run counter to the ideas discussed in Chapter 5 about the possibility of environmental modernisation and the green economy to frame a capitalist growth-based route to sustainability. Indeed, the idea of the 'green economy' (Chapter 5) gives a special place to business corporations as part of a transition to sustainability (Hawken *et al.* 1999).

Notwithstanding arguments about capitalism's 'capture' of environmental concerns through market-based environmentalism, the dominant relationship between environmentalists and corporations since the 1970s has long been antagonistic. Environmental groups developed tactics of direct action to attract media coverage and highlight the damage to the environment caused by corporations. Iconic examples include the action by Friends of the Earth UK against the decision in 1971 by the soft drink company Schweppes to use non-returnable glass bottles (dumping 1,500 bottles outside their headquarters), or Greenpeace's occupation of Shell's redundant Brent Spar oil platform as it was dragged from the North Sea towards the Atlantic to be dumped at sea in the 1990s (Rose 1998): Greenpeace believed it should be recycled, as it eventually was. The tactics were crude, confrontational and designed to attract public attention. To many environmentalists, such actions were inspirational. To executives within global corporations, such activities made environmental groups enemies, 'ecoradicals' driven 'more by hostility toward capitalism than concern for nature' (Lewis 1992, p. 182).

The critical element of corporate campaigning was to use media coverage of actions to highlight a campaign that called for change. By the 1980s, an increasingly important dimension was to influence consumers, asking them to boycott particular products from targeted companies (Scales 2014a). The logic was that actual and feared reductions in sales would force companies to change production methods.

Plate 6.1 Friends of the Earth protest about non-returnable bottles, 1971

In a classic early example of direct environmentalist action, Friends of the Earth campaigners left hundreds of non-returnable bottles on the doorstep of the head office of the soft drink company Schweppes in Marble Arch, London, on 30 October 1971. Schweppes had recently shifted from returnable to single-use glass bottles, part of the whole drinks industry's switch to single-use containers, with glass soon to be replaced with plastics.

Photo: Getty Images.

Thus the Friends of the Earth Rainforest Campaign, launched in 1985, involved a boycott on high-street outlets using tropical timber obtained through clear felling or non-replacement selective logging. As such, campaigns grew in scope and sophistication; environmental campaigners also bought shares in companies and spoke out at shareholders' meetings about environmental and human rights performance. Such publicity increased pressure on chief executives and board members to avoid arguments and to reassure corporate investors worried about the reputation and the future profitability of the company. It might be said, too, that some investors and corporate executives also acknowledged the need to take sustainability seriously, especially as the generation exposed to environmentalism through school and university began to enter positions of authority.

The 1990s saw a change in the simplistic tactics of crude opposition of environmental groups to the actions of corporations. Greenpeace, for example, advocated 'solutions-led' campaigning. This required research and development of alternative technologies and then a campaign directed towards making businesses adopt them. One example of such an approach was the development of propane refrigeration technology as an alternative to the use of CFCs or HCFCs in the fight against the ozone hole (Rawcliffe 1998).

The 1990s also saw the expansion of direct discussions and even collaborations between environmental groups and corporations in a number of sectors, discussed further below. The strategies of environmental groups diversified to embrace the search for influence through discussion and direct meetings, the use of

bold and eye-catching public protests, and legal action. Artists and celebrities were increasingly drawn upon for endorsements. Social media became an increasingly important means of developing and organising support; for example, in the Friends of the Earth project to spell out the words 'climate', 'justice' and 'peace' from linked smartphone locations across Paris during the climate change negotiations in December 2015 (Friends of the Earth 2019). The environmental movement continuously expanded and differentiated as new organisations were founded, often finding and motivating substantial new audiences (for example, Extinction Rebellion, which emerged in the UK in 2018) or the mass climate change strikes of school children in 2019.

6.2.2 The 'green' corporation

During the same period, corporate environmentalism, actions taken by firms to make a substantive or symbolic commitment to ecological protection, began to become more common and more important in both boardrooms and in corporate hierarchies (Mason 2005). The possibility that corporations could become a source of positive social and environmental impact and could deliver wellbeing and sustainability throughout their supply chains, from the way they source raw materials to the way their products and services are designed, marketed and sold, has been an idea much discussed at business schools, by campaigners, and among some chief executives and corporate boards (Hawken *et al.* 1999, Hardyment 2018).

This engagement with the idea that corporations could become 'green' reflected pressure from shareholder calls for environmental and social responsibility – often as part of a more general engagement with corporate social responsibility (CSR). It represented a rethinking of corporate strategy that emphasised the importance of 'brand' values in competition for markets and investment. It also helped position corporations as 'green' or 'caring' to customers. The move was also important internally, as corporate employees began to respond to wider debates about the environment and human rights and wished to feel they worked for a corporation with clean hands and a good heart: the CSR performance of companies was, for example, said to be important in the recruitment of graduates from elite universities. CSR, including human rights and environmental performance, became part of the curriculum of business schools, yet another field within which corporations could seek to separate themselves from rivals, and ambitious executives could advertise their exceptional skill-sets. Such changes reflected the effectiveness of decades of public campaigning by environmental organisations in industrialised countries from the 1960s (Mason 2005).

Awareness of the problem of anthropogenic climate change led to the creation of a transnational advocacy network focused on divestment (persuading financial funds to sell stocks in fossil fuel or other corporations, Ayling and Gunningham 2015). This movement, drawing in elements from the environmental and the poverty and human rights movements, sought to use the power of capitalism to drive corporate greening. It used a range of campaigning strategies to shame, pressure, facilitate and encourage investors (and especially large institutional investors such as pension funds) to reduce or sell holdings of fossil fuel stocks in favour of climate-friendly alternatives. The aim was to restrict the amount of capital available to fossil-fuel companies to prevent them moving to high-cost and high-risk production locations (such as the Arctic, being opened up as polar sea ice melted due to global heating) and to make alternative power generation strategies such as wind or solar relatively more attractive, in the hope of increasing production and lowering unit cost.

The conventional argument about corporate environmentalism is that if environmental performance is something that differentiates companies in terms of both consumer choice and investor confidence, normal processes of competition between businesses can become a driver of corporate change towards more sustainable forms of production and operation (Murphy and Bendell 1997). It is a basic tenet of ecological modernisation that it pays for business to become cleaner because pollution suggests inefficient processes, and greater efficiency can contribute to profits. One can imagine a benign circle where individual corporations compete to clean up their act in search of leaner production (Dryzek 1995, Vlachou 2004).

Logically, at some point, the earning power of corporations themselves might be compromised by the environmental changes caused by their own production or by wider risks such anthropogenic climate change (Enzensberger 1974). What long-term prospects, for example, would lie in leading an oil company in a post-carbon world? Energy supply might offer a sustainable corporate future, but perhaps coal mining or oil extraction from tar sands would not. Of course, such arguments ignore the addiction of modern societies to coal, natural gas, diesel and petrol, and the scale of infrastructural lock-in and subsidy. Feather-bedded by such protection, oil companies remain profitable and their shares remain sought-after, to the chagrin of 'divestment' programmes and climate change campaigners (see Ayling and Gunningham 2015).

However, corporate 'greening' is not driven to any large degree by the moral agendas of individual corporate leaders. Indeed, it is often severely constrained by calculations of value and profit from board members and financial investors, who choose to maximise short-term gain. Yet some kind of environmental action has increasingly proved necessary even for such conservative interests. One driver for this is government regulation. Such regulation in industrialised countries, backed by environmentalist pressure, has effectively outlawed the once common practice of externalising costs by releasing waste unprocessed into a nearby river or the sea. Likewise, the classic strategy of moving operations to less tightly regulated countries has been less viable as regulations have become tighter everywhere, and consumers have started to worry about how the things they buy are produced (see below), making dirty production untenable, particularly for those companies large enough to have a global reputation to defend.

In such circumstances, self-regulation by a whole industrial sector becomes an attractive strategy. Motives are often defensive: Matten (2004) notes that corporations often engage in self-regulation to avoid stricter imposed regulation. The costs of meeting regulations (for example, by retro-fitting pollution-control technology to a factory) can be high and their timing difficult to predict. Such costs and uncertainties can have effects on corporate performance. An ability to predict these costs can therefore be useful. Under these circumstances, direct engagement by corporations in moves towards environmental regulation can make sense, allowing them to influence the speed of change specified in legislation and steer it in a convenient direction. Indeed, governments sometimes explicitly threaten enforced regulation in the hope that self-regulation will be offered instead (Matten 2004). Self-regulation and voluntary 'greening' can help provide a level and affordable playing field for innovation, can make it easier to organise investments in a timely and predictable way, and can also help limit public and government demands for greater environmental accountability and regulation.

Moreover, an innovative corporation can win a competitive advantage over rivals. If a move to a more sustainable form of production is inevitable (for example, fitting pollution-control equipment, or shifting to a less environmentally damaging product), then it is in the interest of individual corporations to make that move in a way that gives them a competitive advantage. This advantage might come through the timing of the change (for example, when they are about to commission a new plant), or by influencing the technical requirements (for example, to use a process for which they own the patent). State environmental regulation can favour capital-rich and technologically strong companies within industrialised economies and provide a means to establish *de facto* trade barriers against corporate competitors, particularly in the developing world.

Environmental performance and sustainability have certainly become an important field for competition with rival enterprises. Responding to opportunities of this kind, leading companies in a succession of sectors took a deliberate step away from rivals in the 1990s and 2000s to try to establish a 'green' brand and to claim emerging environmentally conscious markets. Examples include BP in the oil and gas sector (using its slogan 'beyond petroleum' in a brand re-launch following its merger with Amoco in 1998), or the car manufacturer Toyota launching the hybrid (petrol/electric) Prius in 1997. Such moves, of course, are vulnerable if events suggest environmental claims are bogus (mere 'greenwashing'), as for example BP discovered after the Gulf of Mexico fire of 2010, which gave the lie to their claims about good environmental performance.

Whatever the motivations, there is no doubt that the environment has become a significant issue for many corporations. For example, in 1995 the managing director of the UK hardware chain B&Q wrote: 'we recognised that our stakeholders, that is our customers, staff, local communities, young people, government and shareholders, believed that the environment did matter' (B&Q 1995, p. 3). From the 2000s, corporations began routinely competing on the environmental quality or performance of their products, be it dolphin-friendly tuna, organic yoghurt or 'green' cleaning products. Such claims tend to be selective, indeed often no more than superficial, but they do mark a significant change in corporate rhetoric at least. Such products, and such advertising strategies, have become routine, with those designing them seemingly immune to irony. As global car companies responded in the 2010s to regulations aimed at improving urban air quality and encouraging a shift away from fossil fuels, their advertisements turned on (often exaggerated) claims of 'green' credentials for hybrid or electric vehicles. Thus an advert for a BMW hybrid car was banned in the UK in 2017 because the maker's claim that it was a 'clean' car with zero emissions that helped 'to give back to the environment' was untrue because it had a petrol engine (Harvey 2017).

The Rio Conference in 1992 was an important landmark in the development of corporate engagement in sustainability issues. Corporations funded a fifth of the costs of the conference secretariat and parts of the programme (Chatterjee and Finger 1994). In 1990, the Swiss businessman Stephen Schmidheiny founded the World Business Council for Sustainable Development (WBCSD), following an initiative by Maurice Strong, Secretary General of the Rio Conference (Timberlake 2006). In June 1992, their report on environment and development, *Changing Course*, was published in time to feed into the Rio process (Schmidheiny 1992). The conference and the WBCSD provided a platform for international business to present its view of environment and development, and indeed to present itself as a central part of the solution (Welford and Starkey 1996, Utting 2002).

There was widespread corporate adoption of environmental management systems (notably the International Standards organization's ISO 14001 and the European Union's Eco-Management and Audit Scheme (EMAS)), as corporations have sought to reassure critics and investors that they have moved beyond minimalist compliance with environmental regulations (Mason 2005). The boardrooms of public companies and business schools adopted the language of corporate social responsibility, both as a matter of duty (as a 'responsible corporation') and commercial competitiveness (as annual shareholder meetings provide a public platform where chief executives and board chairs could be embarrassed). Issues of human rights and environment have moved towards the centre of corporate concerns, although such commitments can be much easier to draft than to implement, as BP's experiences in the Gulf of Mexico or Shell's experiences in the Niger Delta in Nigeria attested (Ite 2004, Safina 2011).

Agenda 21 called for partnerships between environmental groups and businesses, and the development of such relationships has been an important element in the post-Rio agenda. There have also been many experiments with partnerships between businesses and NGOs (Murphy and Bendell 2002). Such alliances fit the wider context of market environmentalism, claiming to deliver improved environmental policy while avoiding the unwelcome burden on private enterprise of state bureaucratic regulation (Arts 2002).

The idea of a green economy (discussed in Chapter 5) offered unprecedented opportunities for business corporations. The president of the World Business Council for Sustainable Development said in 2012,

> a capitalist is somebody who optimizes returns from capital employed. The mistake is that the current economic model is solely focused on optimizing the return on financial capital. We need to add two more elements of capital: natural capital and social capital, and tell capitalists to go and optimize that.
>
> (Bakker 2012)

It is the prospect of win–win outcomes, including secure future profit, that motivates moves towards corporate sustainability.

Some chief executives and boards may have been persuaded of the moral imperative to take environmental aspects of their business seriously. Others have seen marketing and growth opportunities in realigning their business to meet the new demands of consumers and the new opportunities in 'green' products. In this formulation, environmental care can simply make good business. In principle, such businesses may promote sustainability, although not all aspects of 'green' technologies are benign. Solar panels and 'smart' digital devices depend on rare earth elements and minerals like coltan, creating unregulated mining booms in countries like the Democratic Republic of Congo (Jeffries 2015). Phenomena like Internet shopping may seem dematerialised to consumers, but it depends on massive energy consumption by server farms.

Much corporate greening is little more than skin-deep, and many environmentalists reject environmental claims as hype and speak of attempts to 'greenwash' corporate operations using very narrowly selected criteria of environmental improvement. The specific commitments of individual manufacturers or traders have sometimes been shown up by catastrophic performance. The vertical integration of commodity chains has made effective oversight of the environmental and social conditions of production by consumer or regulator effectively impossible (Lebel 2005). Nonetheless, since 1992, corporations have claimed a significant role in the delivery of global sustainability. The WBCSD has about 200 members from more than 35 countries and 20 major industrial sectors, involving over 1,000 business leaders. It remains a significant global actor, considerably less conservative than some governments in discussing and calling for change in fields such as energy (WBCSD 2005).

In 2007, the United Nations Secretary General (Ban Ki-moon) launched an initiative called *Caring for Climate*, to create a cadre of business leaders to lead global corporate responses to climate change. The focus was on the efficiency of energy usage, the carbon burden of products and the search for comprehensive, long-term and effective legislative and fiscal frameworks that might make markets work for the climate by creating a market for carbon (Gallager 2016). Corporate executives whose organisations had become 'Carbon Pricing Champions' under the *Caring for Climate* initiative held a side meeting at the 21st Conference of the Parties of the UN Framework Convention on Climate Change in Paris in 2015 to advocate for a global price on carbon, and they engaged with government and civil society leaders to promote a critical role for the private sector in confronting climate change (Gallager 2016).

As discussed in Chapter 5, market environmentalism holds that the world can literally grow out of global environment and development problems and that consumption can be the engine through which sustainable environments and livelihoods are to be achieved. To business advocates of market environmentalism, it is the corporation and the consumer that can deliver on the mainstream agenda of sustainable development. Capitalist markets can clean up and 'green' the planet (Low and Gleeson 1998, p. 81). Thus Martyn Lewis (1992, p. 183) concludes that, 'regardless of extremist fantasies, we can expect that once capitalist energies begin to be harnessed to environmental protection, a virtuous spiral will begin to develop'. A sustainable environment is to be achieved through self-regulation (not heavy-handed regulation by the state), through a revolution in corporate thinking that 'greens' industry from within, combined with a hard-headed (and increasingly global) pursuit of the 'green' consumer dollar.

In visions of corporate-led transitions to sustainability, there is still a role for the state. Basically, the state needs to do two things. First, it needs to regulate. The market may be efficient, but it needs management, both to create a level playing field between competitors to prevent a simple 'race to the bottom' and to drive innovation in technology and institutions. Second, the state has to tax to make the market tell the ecological truth (to reflect costs of resource use), to drive positive change, and to shape consumer behaviour (nudging or shoving to promote less environmentally damaging forms of consumption). Third, the state has to pick up the pieces, paying the costs of unforeseen or residual risks and liabilities that corporations cannot meet because of bankruptcy or voluntary liquidation.

Environmentalists tend to doubt the possibility of distinguishing between genuine changes in corporate activity (perhaps as part of some kind of shift to a real 'green economy', UNEP 2011, Brockington 2012) and simply changes in stories about behaviour (some kind of superficial 'greenwashing' of unchanged

activities, Mulligan 1999, Lyon and Maxwell 2011). Industries, particularly the extractive industries, have been seen by environmentalists as the drivers of environmental destruction for the last century and a half, since the birth of the modern environmental movement. They still occupy the top spot: as the veteran conservationist Michael Soulé (2013, p. 896) notes, 'globalization of intensive economic activity has accelerated the frenzied rush for energy and raw materials and is devouring the last remnants of the wild, largely to serve the expanding, affluent, consumer classes in industrialized and developing nations'. The particular case of the mining industry is discussed later in this chapter, in section 6.4.

The World Economic Forum is a key arena within which new narratives of corporate global citizenship are forged and disseminated. The 48th Annual Meeting, held in January 2018, described itself as 'a true summit of summits'. In the hope of 'creating a shared future in a fractured world', its purpose was 'to rededicate leaders from all walks of life to developing a shared narrative to improve the state of the world', in an 'unparalleled global effort in co-design, co-creation and collaboration'. This vision of a corporate-led path to sustainable development continues to win passionate believers.

6.3 Greening Consumption

6.3.1 Linking production and consumption

The decisions of corporations about their products constitute one side of sustainable production and consumption. The other is the outcome of the decisions of consumers. The environmental and social impacts of production and consumption, and the supply chains that stretch across the globe, are recognised as an increasingly important factor in sustainability. Environmentalists often express severe doubts about the possibility of a society 'buying its way to sustainability', and there is increasingly sharp debate about the validity and utility of the concept of 'sustainable consumption' (Jackson 2006). Yet consumers are a key target for environmentalists calling for wise choices about purchases. And global brands, particularly in sectors such as fashion, have become sensitive to criticism of their products and the conditions under which they are produced.

Supply chains linking producers and consumers have become important to questions of sustainability for several reasons. First, corporations fear damaging headlines in key markets and the attentions of campaigners and media using the global reach of digital communications; second, supply chains have become the focus of regulators in industrialised countries – for NGOs campaigning on issues ranging from child labour to land management; and third, corporations are concerned to secure supply chains against the risk of disruption from resource shortages or extreme events (O'Rourke 2014).

The desire to develop visibility along supply chains and to regulate work practices and environmental impacts at every stage of materials supply and manufacturing has stimulated development of a variety of tools and approaches to enable quality-management systems to take account of sustainability in sourcing, design and transportation. Critical issues include consumption of water, energy and waste across production, shipping and retail, and labour conditions under which products are made and moved. Approaches include indicators, eco-certification, Life Cycle Assessment, Material Flow Analysis (MFA) and footprinting (Finnveden *et al.* 2009).

Life Cycle Assessment (LCA), for example, is an approach to the assessment of impacts of a product throughout its life cycle (raw material extraction, manufacture, transportation and sale, use and disposal, Hellweg and Canals 2014, Hauschild *et al.* 2018). LCA fits relatively easily into corporate planning if applied early in product design (for example, in the parameters for a novel product such as a 'green car': how much post-consumer waste material is to be re-used, how much waste may be generated in manufacture, and what proportion of the vehicle can be recycled once obsolete using existing public facilities). However, LCA can also be applied retrospectively and far beyond an individual product – that is, across the whole production system of one corporation, through its networks of suppliers and clients, and even at the national scale (Hellweg and Canals 2014).

The idea of the 'circular economy' is a model for the continuous flow of technical and biological materials around a 'value circle' from production to consumption and back again via linked waste/raw material flows (Ellen MacArthur Foundation 2018). The same concept lies behind 'Factor 10', a circular economy project of the World Business Council for Sustainable Development (WBCSD 2019).

Many corporations have developed initiatives to address supply chain impacts. In the clothing industry, for example, corporations such as Puma, Levi's, Walmart and Patagonia have initiatives to address different parts of the whole chain from production to consumption (agricultural production of fibre, spinning, weaving and dying, cutting, sewing and finishing, distribution, retail, use and eventual disposal, O'Rourke 2014).

The speed and flexibility of modern supply systems has radically shortened the period between production and consumption for many leading corporations. This does not make supply chain visibility easy to achieve. Fluid and adaptive management of suppliers and minimisation of warehoused stock have made it hard to keep track of where products are sourced, or the conditions under which they are made. Digital technologies have come to play an essential role in tracing products through supply chains, as has acceptance that the challenge is one of creating integrated supply systems rather than discrete chains (Dooley 2014). Standardisation of products, modular shipping containers, more complete utilisation of

Plate 6.2 A container ship in the South China Sea

Supply chain logistics have been transformed by the use of standard shipping containers and computer systems to optimise logistic planning and freight routing. Companies minimise warehouse stocks of components and finished products, and multiple supplier contracts are used to ensure on-time delivery with minimum delay and waste. The complexity and flexibility of such systems can in practice make it hard to understand let alone manage sustainability along supply chains. Here the *Gunhilde Maersk* carries cargo from a container port in Vietnam across the South China Sea, one hop in the seamless web of global production and consumption.

Photo: Toby Smith.

transportation and storage facilities and optimisation of routing all have a role in improving the sustainability of production and supply. Those concerned with such improvements emphasise the importance of attention to the sustainability of each individual element of production and distribution and consumption (Mervis 2014).

Recognition of the importance to sustainability of the links between production and consumption has thrown attention on the need to analyse material and energy flows. The notion of humanity's 'environmental footprint' (Hoekstra and Wiedmann 2014) has drawn attention to the links between consumption in one jurisdiction (e.g. in a developed country) and production somewhere else. International trade has, for example, been shown to be a major cause of biodiversity loss: analysis of 25,000 animal species and 15,000 commodities produced in 187 countries showed that 30 per cent of global species threats were linked to identifiable supply chains between rich consumers and developing-country suppliers (Lenzen *et al.* 2012).

It is widely recognised that globalisation allows unsustainable production practices to simply be moved offshore, but the relationship between domestic and overseas production and consumption is highly complex: diverse, interlinked, multi-staged and, to some extent at least, within the firewalls of private corporations.

The complexity of systems of production and consumption make the analysis of supply chains challenging. Garmendia *et al.* (2016) use the methods of material flow accounting to analyse the material and energy flows of the historically affluent and open economy of the Basque Country. The resulting assessment of the social metabolism of the Basque economy highlights both the dependence of domestic consumption on key commodity flows (such as ocean fish and tin) and the impacts of that consumption.

Most footprint analyses do not provide spatially explicit data across the production-to-consumption system, and this can lead to misleading generalisations (Godar *et al.* 2015). The impacts of production reflect factors such as soil, climate, economy, infrastructure and agricultural technology, and they may be very localised. Spatially explicit analysis of production data can be achieved by combining detailed production data at sub-national scales (municipalities or provinces) with information on domestic flows of goods and in international trade (Godar *et al.* 2015). This kind of analysis allowed identification of the 1,832 specific municipalities (out of 5,565 in Brazil) from which different countries (China, EU, Nordic countries and Brazil itself) sourced the Brazilian soy they consumed in 2011 (Godar *et al.* 2015).

A development of this approach called the *Trase* initiative ('transparent supply chains for sustainable economies', led by the Stockholm Environment Institute) collates data from the FAO trade data, customs authorities, sub-national taxation authorities, bills of lading, exporters, and storage and processing facilities to link the production of soya in specific landscapes in Latin America to the countries where it is consumed (*Trase* 2018a, see Figure 6.1). Its aim is to provide an accessible supply chain information system for over 70 per cent of the total traded volume of the commodities most important to forest loss (soya, beef, palm oil, timber, pulp and paper, coffee, cocoa and aquaculture). The initial focus of *Trase* is on soya across Latin America, beef in Argentina, Brazil and Paraguay, palm oil in Indonesia and Colombia and coffee in Colombia (*Trase* 2018b). There is evidence that public policy interventions on supply chains can be effective. In Brazil, Amazonian deforestation declined by 70 per cent between 2005 and 2013 (19,500 km^2 per year to 5,843km^2 per year). A major factor was continued pressure to limit forest removal by soya and beef producers (Nepstad *et al.* 2014).

6.3.2 Certification schemes

Since the end of the twentieth century, a key component of attempts to 'green' production and consumption has been the creation of voluntary 'sustainability standards' for commodity production, widely referred to as certification schemes. A wide range of such schemes exist, addressing both environmental and social dimensions of production.

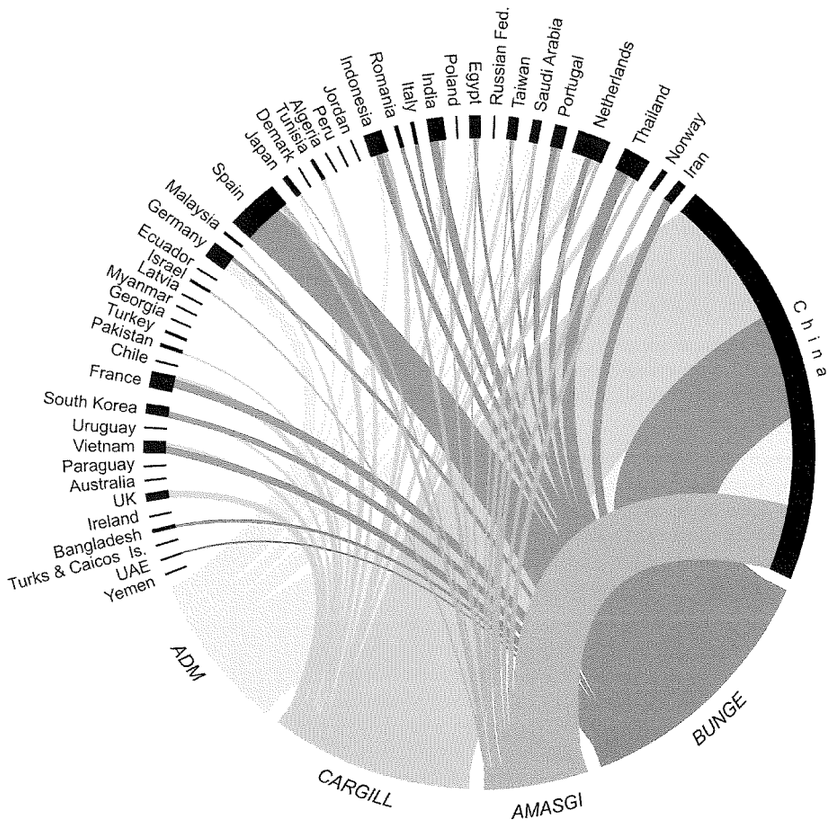

Figure 6.1 Exports of soy in 2015 from Brazil

The *Trase* project ('transparent supply chains for sustainable economies'), led by the Stockholm Environment Institute, collates official data to link the production of soya in specific landscapes in Latin America to the countries where it is consumed, via traders and processors. This diagram shows the destination of exports of soya from Brazil in 2015 through four leading international produce companies.

Source: Stockholm Environment Institute (https://trase.earth/?).

Three 'fair trade' labels were introduced in Europe in the late 1980s (Max Havelaar, Fairtrade Mark and TransFair), soon extending to the USA, Canada and Japan (Raynolds *et al.* 2004). The UK NGO the Fair Trade Foundation was established in 1992 by CAFOD, Christian Aid, Oxfam, Traidcraft, Global Justice Now and the National Federation of Women's Institutes (Fair Trade Foundation 2019). It is part of a much larger network of organisations globally, in Fairtrade International, which links labelling and certification initiatives across Europe, Japan, North America, Mexico and Australia/New Zealand as well as networks of producer organisations in Africa, Latin America, the Caribbean and Asia. By 2005, the value of Fair Trade products globally was more than 5 billion Euros (Le Velly 2007). In 2019, the Fair Trade Foundation claimed that 1.66 million farmers and workers were in 1,411 Fairtrade-certified producer organisations worldwide (23% of them women). In 2016, the Fairtrade premium paid was 158m Euros (Fair Trade Foundation 2019).

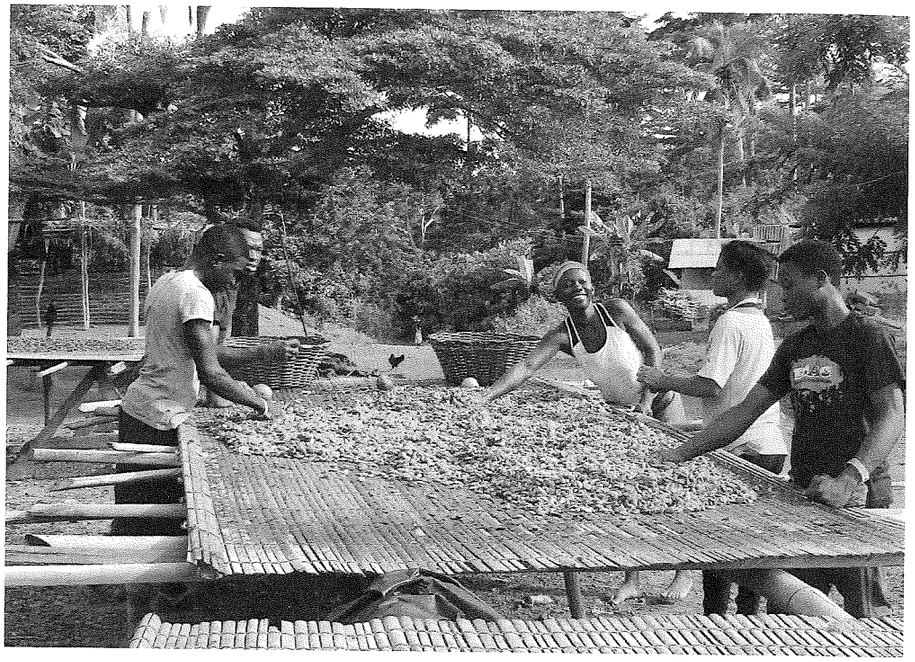

Plate 6.3 Smallholder cocoa drying in Ghana

Cocoa beans being dried after fermentation in a small village in southern Ghana. Drying over five to seven days allows acids in the cocoa to evaporate, yielding a good quality bean with low acid levels and good flavour. After drying, beans are sorted, bagged and sold to buyers. Ghana is the second largest exporter of cocoa (behind Ivory Coast), and it is Ghana's biggest cash crop. Large chocolate manufacturers such as the Swiss transnational food and drink company Nestlé have started to source Fairtrade cocoa for some products and in some markets (e.g. the UK and Ireland, https://www.nestle.co.uk/media/pressreleases/100pc-sustainable-cocoa). Specialist artisanal chocolate manufacturers often deal directly with small-scale village cooperatives (e.g. http://clonakiltychocolate.com/about/ghana/).

Photo: Bill Adams.

Key 'fairtrade' products are coffee, tea and cocoa. These are all long-standing global commodity crops whose trade is dominated by a small number of large international companies that are vertically integrated (from plantation to cup) and that sell using internationally recognised brands. The development of 'fairtrade' products demands direct competition with these established global-scale corporate producers.

Fair trade and organic production emerged from social movements seeking an alternative to mainstream international commodity markets that have long been highly centralised and controlled by a few large globally integrated businesses. The Fair Trade movement seeks to address global poverty and inequality by offering a reformed system of trade that empowers small-scale Southern producers by providing direct access to ethically minded Northern consumers. The notion of 'fair' trade involves issues of fairness in produce pricing and questions of working, employment and living conditions, and environmental sustainability. The Fair Trade Foundation, for example, sets social, economic and environmental standards for companies (including payment of a 'Fairtrade Minimum Price' and an additional premium for investment in projects of the community's choice) and farmers (including protection of workers' rights

and the environment). These standards are checked, and the 'fairtrade' mark on raw materials and products provides a guarantee through the supply chain, from field to supermarket shelf (Fair Trade Foundation 2019).

Many certification schemes have emerged from direct collaborations between commercial producers and their critics. One example is the work of the Round Table on Responsible Soy (RTRS). This was set up in 2006 by Grupo Maggi, Cordaid, COOP, WWF, Fetrauf-Sul and Unilever. A sustainable soy standard was agreed in 2010, and the first producers were certified in 2011 in Argentina, Brazil and Paraguay (RTRS 2019). Certified soya production has to meet stipulated standards of environmental management, agricultural practices, labour conditions, community relations and legal and business practice – the main standard being to promote 'responsible production, processing and trading of soy on a global level' (RTRS 2019). It does not include the clearance of forests in the first place to clear land for soy production.

Certification schemes relate not only to fair trade but to the state of nature. In the context of tropical forests, the US-based NGO Rainforest Alliance certified its first coffee in 1990 (Rainforest Alliance 2019), and the giant US Starbucks coffee chain claims that 99 per cent of their coffee is accredited by US conservation NGO Conservation International under a programme that began in 2004. The rise of Fair Trade and other forms of certification was given a huge boost by the collapse in green coffee prices between 1999 and 2004, which brought devastating hardship to coffee-producing communities. Thus in Central America, amid humanitarian and food security emergencies, collaboration between Nicaraguan smallholder cooperatives and small Californian roasting companies led to alternative models of coffee supply, linking cooperative producers and Bay Area coffee drinkers (Bacon 2013). This account of certification identifies individual consumers as the key actors in a transformation of consumption reflecting a revolution in ideas about product quality, environment and ethics.

In the marine environment, Unilever (the largest global seafood corporation with brands such as Birds Eye) and the World Wide Fund for Nature (WWF) formed the Marine Stewardship Council (MSC) in 1997. MSC certification focused on the individual 'fisheries', addressing the state of fish stocks, the impacts of fish capture (e.g. trawling) and the rules and procedures in place to ensure sustainability (Cummins 2004). By 2018, MSC claimed that 14 per cent of the global marine catch came from fisheries that were either certified or engaged in the MSC programme (MSC 2018). They were one of a series of global and national schemes to certify the sustainability of commercial fisheries. Their approach involved recognising and rewarding sustainable fisheries, incentivising improvement and cultivating and expanding the market for sustainable seafood through engagement with key commercial partners, and working especially in Europe (Germany, the UK, France, Spain and Italy), the US, China and Japan, and building public awareness and support for sustainable fishing and for their own MSC 'ecolabel' brand (MSC 2018).

But the possibility of a consumer-led revolution of this kind is widely questioned and with it the whole notion of 'sustainable consumption' (Zaccaï 2007). The very success of certification in winning a slice of commercial trade throws up a challenge: if the same tools that drive conventional consumption are used to build up the market share of certified products, can the new way of doing business really be different from capitalism's entrenched system (Le Velly 2007)? The centrality of advertising and brand development to certified product chains has been widely remarked upon. The emergence of speciality and certified coffees implies that different dimensions of consumerism can be combined: economic choice (about price, quality or convenience), political action (the idea of community-of-sentiment) and identity production (how purchasers see themselves as people in the world, not just as 'consumers', West 2010).

This process involves what Karl Marx described as the 'fetishism' of commodities under capitalism, meaning their presentation to purchasers in ways that ignore or deny the labour invested in their production (Carrier 2010). Commodities – like packets of coffee – exist as simple objects, products on a shelf, delivered down a commodity chain that hides the conditions of production. It is precisely this on which certification proposes to shine a light, making the conditions of production legible. Yet the very creation of a certified commodity – the stories told on packets about hard-working farmers, the images of

cloud forests or smiling cooperative members – involves the presentation of a highly stylised picture of a self-reliant peasantry in a sustainable landscape. Wright (2004) analysed the representations of 'authentic' lives and landscapes in an early newspaper advertising campaign (1999–2002) for UK Cafédirect (founded in 1991 by Oxfam Trading). Striking imagery and related text addressed social relations behind the production, exchange and consumption of coffee, but the story told was so limited and idealised that producers and landscapes were effectively being enrolled as part of the product in just the same way as conventional branding.

The making of any product, even a certified product like coffee or cocoa, involves the creation of such disembodied commodities, a process Marxists call 'fetishisation'. The actual conditions of production – the politics of cooperatives, issues of land rights, the chain of traders, roasters, shippers, wholesalers and retailers – are hidden from view behind packet, trademark and associated images, just as they are for conventional coffee (Carrier 2010). Paige West (2010, p. 692) reminds us that behind the 'veritable love fest of learning about each other, supporting each other, and contributing to environmental sustainability', speciality coffee is still neoliberal coffee: it has found a place in the market because of changes in the way markets are organised, global in reach, highly individualised and tightly linked to image and brand. Ethical consumers are reached, and effectively created, by advertising as a normal part of commercial market segmentation. Drawing on ethnographic work, West describes the huge distances between the lived reality of coffee production in Papua New Guinea and the images of coffee producers around whom the new coffees are sold: 'fantasy images of ecologically noble savages and pure guileless ecological primitives' are careful productions and they have material consequences (West 2010, p. 710).

The crunch question for certified commodity chains must be what difference they make on the ground. And the answer to this is, inevitably, that it depends. It depends on the nature of local institutions – which people are allowed to become members of – cooperatives or other producer organisations (in terms of gender, ethnicity or wealth, for example), how many people in the community depend on employment as labourers and how they are treated. It depends on the equality of land distribution and how secure land tenure is. It depends on the continuity and predictability of commodity prices, on the extent to which mark-ups are added through the consumption chain. It depends on the costs of certification (usually borne by producers) and the fairness of the process – certification schemes depend on local agents to carry out inspections, and such arrangements face all the usual issues of competence and corruption.

In the case of environmental certification, a global map synthesising data from over a million farms showed a broad concentration in areas of high biodiversity, with a less good fit for areas most in need of poverty alleviation (Tayleur *et al.* 2018). Yet the critical issue is effectiveness in terms of the extent to which producers change their land management. In the Western Ghats of India, for example, coffee growers were provided price incentives to adopt Rainforest Alliance certification standards. Yet certification did not change the way land was managed, since coffee was already shade-grown and met the other required standards. Certified growers were disillusioned because of the lack of constraint on their practices. Although labelled as a conservation initiative, growers saw certification as a market tool to serve the needs of coffee traders and exporters (Bose *et al.* 2016).

Of course, it is easy to attack certification for its failings and limitations. The self-righteousness of middle-class consumers, piously proposing to right the world's wrongs one flat white at a time, is catnip to radical critics. Like other forms of 'ethical consumption' in food (e.g. the concern for food miles, the rise of 'locavore' lifestyles – eating only what is produced locally – the notion of 'slow food' or the rise of farmers' markets), a strong backlash has arisen among both researchers and popular writers (Schnell 2013).

There is also a tendency for corporations to move away from well-established independent certification schemes. Many large corporate retailers in the UK are abandoning established schemes such as Fair Trade in favour of bespoke arrangements, blending any charitable aims more seamlessly with their corporate brand. This kind of 'private eco-branding' allows retailers to govern sustainability dimensions of contractual relationships in the value chain, but it also allows them to capture gains from investments in

the development of sustainable supply chains in terms of brand value and customer loyalty (Chkanikova and Lehner 2015).

Certification may not live up to all the expectations laid on it. Yet it certainly creates a space for humane and equitable relations with producer communities and along supply chains. It is better – often substantially better – than rapacious land and community-destroying commercial practices of the past. But that sets a low bar (West 2010). Collaboration between development and conservation NGOs with businesses to cast light on dark supply chains can achieve small improvements. But it is a marginal tweak to the tuning of capital's engine of production and consumption, not an alternative model.

6.3.3 Regulating timber

One of the earliest attempts at global self-regulation in the face of environmental critique was the tropical timber trade. The complexity and international connectivity of timber markets has long made regulation by individual national governments problematic. Attempts at global coordination of forest management – for example, the Tropical Forestry Action Plan (FAO 1985) – had limited purchase. One reason is that attempts to regulate international trade in timber and forest products have taken place in the era of neoliberalism and 'free trade'. The World Trade Organization has not been friendly to attempts to regulate trade on environmental grounds (Hartwick and Peet 2003). Support for neoliberal strategies (deregulation, privatisation and structural adjustment) has limited the World Bank's capacity to promote sustainable forestry. Organisations like the United Nations Forum on Forests have been described as talking shop without teeth (Humphreys 2006).

A critical challenge for those who seek to establish sustainable forestry through regulation is the problem of illegal logging. This is a major problem in most forest-rich timber-producing countries (Smith 2004a, 2004b). In some countries, more than half the timber being traded was estimated to have been harvested illegally, with up to 70 per cent in Indonesia, 80 per cent for the Brazilian Amazon and 90 per cent in Cambodia (Smith 2004a). Illegally logged timber is used to meet developed world demand for such 'vital' products as hardwood garden furniture (Alley 1999). In so doing, the economic values of the standing forest to local people are liquidated and privatised. Because of its illegality, the benefits of such trade are lost to the public purse and recycled abroad or through elite consumption and the black economy.

In 1983, The International Tropical Timber Organization (ITTO) was established between producing and consuming countries to improve tropical forest management, improve the marketing and distribution of tropical timber and to promote wood processing in the producing countries (Johnson 1985). In 1991, the ITTO set the goal of making the global timber trade sustainable by the year 2000, disseminating guidelines and promoting the case for sustainable management to member governments. However, its progress was slow and achievements very modest (WWF 1996). While the word 'sustainable' was attached to many tropical hardwood products, and claimed for many production systems, the majority of tropical forestry continued to degrade forests and cut timber at unsustainable rates. The problem of 'timber mining' continued.

The drive towards the certification of the origin of tropical timber grew out of campaigns against unsustainable logging by environmental NGOs from the 1980s onwards. Media campaigns achieved considerable traction (in September 1989, the cover of *Time Magazine* showed rainforest brining in the Amazon), and attention turned to consumer boycotts of specific companies and products. In the early 1990s, various environmental groups (Friends of the Earth, rainforest action groups and eventually Earth First!) began direct-action protests against the six largest DIY superstore chains in the UK. However, there was no immediately available alternative to the unacceptable and unsustainably produced tropical timber products – no carrot to combine with the stick of consumer protest. An attempt was therefore made in the 1990s to provide this through certification of the origins of products and the conditions under which they were produced. The idea was that, if Northern consumers could distinguish products produced sustainably from those that were not, their purchasing power could drive production towards more sustainable methods.

In 1991, the WWF UK persuaded ten companies producing or using tropical timber to commit themselves to sustainable forest production by 1995, the '1995 Group' (Murphy and Bendell 1997). The 1995 Group agreed to move towards phasing out non-accredited wood. They adopted the certification system of the Forest Stewardship Council (FSC), established in 1993 as an independent source of certification and labelling of 'sustainable' timber. The FSC is a membership organisation, with both industry and environmental interests represented in two chambers. It adopted ten principles of forest management (see Table 6.1).

The 1995 Group committed themselves to identifying the source of wood and wood products and to appointing a named senior manager to implement this commitment and monitor progress every six months. The companies and the WWF stayed somewhat at arm's length – WWF UK did not verify individual companies' environmental performance, and companies did not gain use of the WWF panda logo. They could use the FSC logo on appropriate products (WWF 1996).

These developments took place against the background of an industry backlash against the partnership and certification process, driven by organisations such as the Canadian Pulp and Paper Association. In 1996, the Timber Trade Federation opposed this arrangement as contrary to the principles of free competition and free movement of goods under UK and EU law, and WWF UK was forced to water down the group's membership requirements (Murphy and Bendell 1997). This was one dimension of a wider and more systematic neoliberal assault on environmentalism in the 1990s, as industries and right-wing political activists in a number of countries (but particularly in the USA) attempted to reverse the

Table 6.1 Forest Stewardship Council principles for responsible forest management

Principle 1:	The Organization shall comply with all applicable laws, regulations and nationally-ratified international treaties, conventions and agreements.
Principle 2:	The Organization shall maintain or enhance the social and economic wellbeing of workers.
Principle 3:	The Organization shall identify and uphold Indigenous Peoples' legal and customary rights of ownership, use and management of land, territories and resources affected by management activities.
Principle 4:	The Organization shall contribute to maintaining or enhancing the social and economic wellbeing of local communities.
Principle 5:	The Organization shall efficiently manage the range of multiple products and services of the Management Unit to maintain or enhance long term economic viability and the range of environmental and social benefits.
Principle 6:	The Organization shall maintain, conserve and/or restore ecosystem services and environmental values of the Management Unit, and shall avoid, repair or mitigate negative environmental impacts.
Principle 7:	The Organization shall have a management plan consistent with its policies and objectives and proportionate to scale, intensity and risks of its management activities. The management plan shall be implemented and kept up to date based on monitoring information in order to promote adaptive management. The associated planning and procedural documentation shall be sufficient to guide staff, inform affected stakeholders and interested stakeholders and to justify management decisions.
Principle 8:	The Organization shall demonstrate that progress towards achieving the management objectives, the impacts of management activities and the condition of the Management Unit, are monitored and evaluated proportionate to the scale, intensity and risk of management activities, in order to implement adaptive management.
Principle 9:	The Organization shall maintain and/or enhance the High Conservation Values in the Management Unit through applying the precautionary approach.
Principle 10:	Management activities conducted by or for The Organization for the Management Unit shall be selected and implemented consistent with The Organization's economic, environmental and social policies and objectives and in compliance with the Principles and Criteria collectively.

Source: Forest Stewardship Council, https://ic.fsc.org/en/what-is-fsc-certification/principles-criteria/fscs-10-principles.